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Econ 475
Fall 2026

Problem Set #1
Due 10/28/2026

Instructions: You may use any statistical software package you choose to complete this problem set. Examples and demonstrations in class will generally use R, but you are free to use another package if you prefer. The solution you submit for each question should be written as a narrative. This includes properly formatted tables rather than copied-and-pasted statistical output, and graphs with informative titles, axis labels, legends where appropriate, and captions. Do not submit only code with comments. In addition to your written answers, submit the code used to generate your results. Your code should be sufficiently organized and documented that another student familiar with the statistical package could reproduce your analysis. You may work in groups of no more than three. In addition to your formal answers, you or your group must prepare a short presentation that represents your solution to each question. As discussed in the syllabus and in class, you may be called upon randomly to present and discuss your solution.

Make sure to label all graphs and tables fully.

Parts marked **[Ph.D. students required]** must be completed by Ph.D. students; undergraduates who complete them well can earn extra credit.

1. Housing Price Indexes

The dataset `price_data.dta` (also used in Problem Set 1) contains 273,661 sales of 241,585 single-family houses in 1,200 census tracts of a major U.S. metropolitan area between 1990 and 2007. About 29,000 houses sell more than once.

A price index answers the question: how much would the same house have sold for in year t compared with the base year? The challenge is that a different set of houses sells each year. In this question you will build several indexes that handle this problem in different ways and compare them. Use 1990 as the base year throughout. Because some houses sell more than once, cluster standard errors by `property_id`.

Log vs. Levels: Each regression below has $\ln p$ on the left-hand side, so the year coefficients $\hat{\delta}_t$ form a *log price index*: $\hat{\delta}_{1990} = 0$, and $\hat{\delta}_t$ is the change in log price between 1990 and year t for a house with fixed characteristics, measured in *log points*. You will first plot the $\hat{\delta}_t$ directly and then learn to convert them into percentage changes and into a conventional *level index* with $1990 = 100$:

$$\%\Delta p_{1990\to t} = 100 \times \left(e^{\hat{\delta}_t} - 1\right), \qquad I_t = 100 \times e^{\hat{\delta}_t}.$$

1. Naive Approach: For each year, compute the mean and the median of `lnprice`. Construct the naive log index $\hat{\delta}_t^{\text{naive}} = \overline{\ln p}_t - \overline{\ln p}_{1990}$ (and the same for the median) and plot it against year. Also compute the mean of `sqft2` and `numbath` for houses sold in each year. Does the mix of houses that sell change over time? Why does that make the naive index misleading?

2. Hedonic index: Estimate

$$\ln p_{ijt} = \alpha + \gamma_1 \,\text{sqft2}_i + \gamma_2 \,\text{numbath}_i + X_j'\beta + \sum_{t=1991}^{2007} \delta_t \, D_t + u_{ijt},$$

where X_j contains the tract racial/ethnic shares and median income, and $D_t = 1$ if the sale occurred in year t (1990 is the omitted year).

   (a) In the lecture, we plotted the constant together with the year coefficients. Explain what $\hat{\alpha}$ and $\hat{\alpha} + \hat{\delta}_t$ measure and why the *difference* $\hat{\delta}_t$ is all we need for an index.

(b) Plot: Plot $\hat{\delta}_t$ against year with a 95% confidence band, setting $\hat{\delta}_{1990} = 0$. Label the vertical axis "log points relative to 1990." (In R, `fixest::iplot()` does this in one line; in Stata, use `coefplot`.)

(c) *From log points to percentage changes.* A common shortcut reads $100 \times \hat{\delta}_t$ as the percentage change since 1990. For 1991 and for 2007, report $\hat{\delta}_t$, the shortcut $100\hat{\delta}_t$, and the exact change $100(e^{\hat{\delta}_t} - 1)$.

- For which year is the shortcut accurate, and for which is it badly off? Is the shortcut too high or too low when $\hat{\delta}_t > 0$?
- Use the approximation $e^x \approx 1 + x$ (a first-order Taylor expansion around $x = 0$) to explain why. Roughly how large can $|\hat{\delta}_t|$ be before the shortcut is off by more than one percentage point?

(d) *From log points to a level index.* Construct $I_t^H = 100\, e^{\hat{\delta}_t}$ (so $I_{1990}^H = 100$) and plot it with a 95% confidence band. Compute the band for δ_t first and then transform its endpoints: $\left[100e^{\hat{\delta}_t - 1.96\,\text{se}},\ 100e^{\hat{\delta}_t + 1.96\,\text{se}}\right]$. Why is this band not symmetric around I_t^H? Interpret I_{2007}^H in words.

3. Replace X_j with a full set of census-tract fixed effects μ_j and save the year coefficients $\hat{\delta}_t^{FE}$. What sources of bias in the hedonic index does this specification remove? What sources of bias remain?

4. Repeat-sales: Estimate the model with a house fixed effect θ_i:

$$\ln p_{it} = \theta_i + \sum_{t=1991}^{2007} \delta_t\, D_t + u_{it},$$

and save the year coefficients $\hat{\delta}_t^{RS}$.

(a) Which houses actually contribute to identifying δ_t in this regression? How many are there?

(b) Compare the average characteristics (size, bathrooms, tract income, tract racial composition) of houses that sell more than once with houses that sell only once. Is the repeat-sales sample representative?

5. Comparisons:

(a) Plot the four log indexes ($\hat{\delta}_t^{\text{naive}}$ using the mean, $\hat{\delta}_t^H$, $\hat{\delta}_t^{FE}$, $\hat{\delta}_t^{RS}$) on one graph.

(b) Make a table that reports, for each index: $\hat{\delta}_{2007}$; the shortcut $100\hat{\delta}_{2007}$; the exact cumulative change $100(e^{\hat{\delta}_{2007}} - 1)$; and the average annual growth rate $100\big(e^{\hat{\delta}_{2007}/17} - 1\big)$. Explain why dividing $\hat{\delta}_{2007}$ by 17 *before* exponentiating gives the annual rate, and why the shortcut works much better for the annual rate than for the cumulative change.

(c) The *difference* between two log indexes is also in log points. By roughly what percentage does the naive index overstate the 2007 price level relative to the repeat-sales index?

(d) Re-plot the four indexes as level indexes ($100e^{\hat{\delta}_t}$). Which version, log or level, would you use to compare growth *rates* across periods? Which would you show to a reader who wants to know how much prices rose? (Hint: in the log plot, a constant growth rate is a straight line.)

(e) Which index grows fastest and which grows slowest? Explain the differences using the ideas of (i) changes in the mix of houses sold, (ii) unobserved house quality, and (iii) selection into the repeat-sales sample (for example, renovations or quick resales by investors).

6. Quarterly index: Re-estimate your preferred model from (b)–(d) with quarter-of-sale dummies built from `yrquarter` (base 1990Q1). Plot the quarterly and annual log indexes together. What do you gain and what do you lose by using a finer time unit? Report the number of sales per quarter to support your answer.

7. Heterogeneity: Split tracts into two groups using a characteristic that varies across neighborhoods, for example, tracts above vs. below the median of `med_income`, or tracts above vs. below the median of `black_share+hisp_share`. Estimate a separate tract fixed-effects (or repeat-sales) log index for each group and plot the two on one graph. Did prices in the two groups move together? By how many percent did each group's prices rise between 1990 and 2007?

8. **[Ph.D. students required]** The Case-Shiller estimator:

   (a) Restrict to houses with exactly two sales, at $s < t$. Show algebraically that differencing the model in (d) gives $\ln p_{it} - \ln p_{is} = \sum_\tau \delta_\tau Z_{i\tau} + \varepsilon_i$, where $Z_{i\tau} = +1$ if $\tau = t$, -1 if $\tau = s$, and 0 otherwise. Estimate this regression and confirm that it reproduces your estimates from (d) for the two-sale sample.

   (b) Case and Shiller (1987) argue that the error variance grows with the time between sales. Implement their three-step weighted repeat-sales estimator: (1) estimate by OLS; (2) regress the squared residuals on a constant and the gap between sales $(t - s)$; (3) re-estimate by WLS using the inverse of the fitted values as weights. Plot the OLS and WLS indexes and discuss.

   (c) Because $\exp(E[\ln p]) \neq E[p]$, the level index $100e^{\hat{\delta}_t}$ tracks the geometric mean price. Briefly explain how this differs from an index that tracks the arithmetic mean price, which is what the S&P Case–Shiller index targets.

========== 2. Chicago Area Price Dynamics:

The DePaul University Institute for Housing Studies (IHS) publishes quarterly repeat-sales price indexes for Cook County, the City of Chicago, suburban Cook County, and 33 submarkets (16 in the city and 17 in the suburbs). The indexes run from 1997Q1 to 2022Q4, with 2000Q1 $= 100$. You are given two files:

- `depaul_q4_price_indices.csv`: the quarterly indexes in *wide* format. There is one row per quarter (`yearq`, e.g., `2008Q3`) and one column per geography.
- `2022_q4_price_index_data.xlsx`: the original IHS workbook. The *Submarket_Definitions* sheet lists the municipalities or community areas in each submarket, and *Submarket_Annual_Sample_Size* reports the number of sales behind each index.

1. Data: Read the `.csv` file into your software.

   (a) Report the number of rows and columns and check that each column was read in with the right type (numbers stored as numbers, not text).

   (b) Convert `yearq` into a date variable that your software treats as time. In R, `zoo::as.yearqtr()` or `lubridate::yq()` will do this.

   (c) Reshape the data from wide to *long* format, with one row per geography-quarter (R: `tidyr::pivot_longer()`; Stata: `reshape long`). Explain why the long format is more convenient for plotting several series with `ggplot`.

2. Plot Cook County, the City, and the Suburbs together on one graph. Then choose *five* submarkets. At least two should be in the city and at least two in the suburbs, and your choices should cover a range of neighborhood incomes. Make a second graph comparing the five submarkets with the Cook County index. Mark the start of the Great Recession (2007Q4) and the start of the COVID-19 pandemic (2020Q1) with vertical lines.

3. For Cook County and each of your five submarkets, compute the following and present the results in a table:

   (a) the boom-era peak (value and quarter) and the percentage change from 2000Q1 to that peak;

   (b) the post-bust trough (value and quarter) and the peak-to-trough percentage decline;

   (c) the percentage recovery from the trough to 2022Q4, and whether 2022Q4 prices had regained the boom-era peak;

   (d) the percentage change from 2020Q1 to 2022Q4.

   Check a few of your numbers against the *Summary* sheet of the Excel file.

4. In 2022Q4, the Englewood/Greater Grand Crossing index is above the Winnetka/Northbrook index. Does this mean houses in Englewood cost more than houses in Winnetka? Explain what an index level of 100 means and what comparisons you can and cannot make across submarkets.

5. For *all* 33 submarkets, find socioeconomic information on the communities measured in 2000 (before the bust). For example, median household income, the homeownership rate, or the share of residents who are Black or Hispanics. Useful sources include the Census/ACS (`https://data.census.gov`), CMAP Community Data Snapshots, or Chicago Data Portal. You will need to aggregate municipalities or community areas into submarkets using the *Submarket_Definitions* sheet. Averages are fine.

   (a) Make scatter plots of the peak-to-trough decline (and, separately, the 2012–2022 recovery) against each characteristic.

   (b) Estimate a simple cross-sectional regression of the decline on your characteristics. With only 33 observations, what do you conclude?

6. Use your results and outside reading to explain the similarities and differences you find.

7. Using the *Submarket_Annual_Sample_Size* sheet, plot the number of sales per year in your five submarkets. Which indexes are based on thin markets, and when? How should this affect how much weight you put on short-run movements in those indexes?

8. **[Ph.D. students required]** For all 33 submarkets, estimate a "beta" by regressing each submarket's quarterly log change on the Cook County quarterly log change. Relate the betas to your neighborhood characteristics from (e).

3. Tenure choice across generations: The dataset `tenure_choice.dta` is an extract of householders from the 2010 American Community Survey (N = 1,190,030). You will use it to study how the probability that a householder owns a home depends on theircharacteristics. The variable `own` equals 1 for owners and 0 for renters. See `tenure_choice_ddf.txt` for the variable definitions. The data are a single cross-section: every observation is from 2010.

1. Variable Construction:

   (a) Compute birth year $= 2010 -$ age and create mutually exclusive cohort dummies (equal to 1 if the condition holds and 0 otherwise):

Cohort	Birth years	Ages in 2010
Greatest/Silent generations	1945 or earlier	65 and older
Baby boomers	1946–1964	46–64
Generation X	1965–1980	30–45
Millennials	1981 or later	29 and younger

   (b) Marital status: create dummies for *married, spouse present*; *previously married* (spouse absent, separated, divorced, widowed); and *never married*.

   (c) Citizenship: create dummies for U.S.-born, born abroad to American parents, naturalized citizen, and non-citizen.

   (d) Income: create the log of `hhincome` and dummies for weighted income quintiles. How many observations have zero or very low income? How do you handle them when taking logs?

   (e) Construct at least two more variables from the remaining data (e.g., employed, self-employed, long commute). Explain why `trantime` $= 0$ for some people and what that means for using it in a regression.

2. Descriptives:

   (a) Make a table means for the full sample and separately for owners and renters. Comment on the biggest differences.

(b) What share of householders is in each cohort?

(c) Describe the income distribution: report the 10th, 25th, 50th, 75th, and 90th percentiles and plot a density of log income. Why is the log scale useful here?

(d) Make a bar chart of the ownership rate by cohort, and a line graph of the ownership rate by single year of age. What does the age graph show that the cohort graph hides?

3. A linear probability model (LPM) is an OLS regression with a binary dependent variable. Its coefficients measure changes in the *probability* that own $= 1$ (see the notes on Blackboard on discrete choice). Use heteroskedasticity-robust standard errors throughout.

   (a) Estimate an LPM with only the cohort dummies, omitting baby boomers:

   $$\text{own}_i = \alpha + \sum_{k \neq \text{boomer}} \psi_k \, \text{cohort}_{ik} + \epsilon_i.$$

   Interpret $\hat{\alpha}$ and each $\hat{\psi}_k$. How do they relate to your bar chart in (b)?

   (b) Add income (log income or the quintile dummies), marital status, veteran status, and citizenship:

   $$\text{own}_i = \alpha + X_i'\delta + \sum_{k \neq \text{boomer}} \psi_k \, \text{cohort}_{ik} + \epsilon_i.$$

   Interpret the coefficients on income and on being married. How do the $\hat{\psi}_k$ change compared with (ii)? Why?

4. Using the model in (c)(iii), predict the probability of owning for a married, U.S.-born householder in each cohort, (1) with household income at the 90th percentile and (2) with income at the 25th percentile. Present the eight predictions in a table. How large is the income gap within each cohort, and how does it compare with the gaps across cohorts? Do any predictions fall outside $[0, 1]$? How many in-sample fitted values do?

5. Logit: Estimate the model in (c)(iii) as a logit: $\Pr(\text{own}_i = 1 \mid X_i) = \Lambda(\alpha + X_i'\delta + \sum_k \psi_k \, \text{cohort}_{ik})$, where $\Lambda(z) = e^z/(1 + e^z)$.

   (a) Why can't you interpret the logit coefficients directly as changes in probability? Interpret the coefficient on being married as a change in the log odds and as an odds ratio.

   (b) Compute *average marginal effects* (R: `marginaleffects::avg_slopes()`; Stata: `margins, dydx(*)`) and put them in a table next to the LPM coefficients. Where are they similar, and where do they differ?

   (c) Repeat the eight predictions from (d) with the logit. For which types of householders do the LPM and logit disagree most, and why?

6. Add interactions between log income and the cohort dummies to the LPM (or logit). Test whether the interactions are jointly zero. Plot the predicted probability of owning against income separately for each cohort, holding the other covariates at representative values. Interpret.

7. Specify your own model (LPM or logit) using any of the variables, explain your choices, and use it to predict the probability of ownership for three different kinds of householders that you describe. Explain what drives the differences in their predicted probabilities.

8. **[Ph.D. students required]** We have been calling the ψ_k "generation" effects.

   (a) In a single cross-section, birth year $c_i = 2010 - a_i$, where a_i is age. Explain why the cohort dummies are a coarse step function of age. Explain why $\hat{\psi}_{\text{millennial}}$ therefore cannot separate a *generational* difference in preferences or constraints from ordinary *life-cycle* effects.

(b) Add a full set of single-year-of-age dummies to the model in (c)(iii) along with the cohort dummies. What happens, and why? Now replace the age dummies with a quadratic (or cubic) in age and keep the cohort dummies. Which variation in the data identifies ψ_k in this specification? Is that identifying assumption credible?

(c) Propose a comparison that would provide more convincing evidence about whether millennials are less likely to own than earlier generations *at the same age* (e.g., comparing 30–34 year-olds across census years). What confounders would remain?

Notes: (1) `age`, `marst`, `citizen`, and the labor-market variables are stored as *labeled* numeric variables. In R, `haven::read_dta()` reads them as `haven_labelled`. Use `as.numeric()` to get the codes or `haven::as_factor()` to get the labels. (2) In `citizen`, the label "N/A" (code 0) means *born in the U.S.* (3) `perwt` is the survey weight. Report weighted descriptive statistics and, unless told otherwise, estimate weighted regressions. (4) If your computer struggles with the full sample, draw a 10% random sample. Set a seed (e.g., `set.seed(475)`) so your results can be reproduced.

Helpful commands

Look up these commands in the help files or online.

R packages: `tidyverse` (`readr`, `dplyr`, `tidyr`, `ggplot2`), `haven`, `readxl`, `fixest`, `modelsummary`, `marginaleffects`, `zoo` or `lubridate`.

Reading data

```
# R
library(haven);  prices  <- read_dta("price_data.dta")
library(readr);  idx     <- read_csv("depaul_q4_price_indices.csv")
# base R alternative:  idx <- read.csv("depaul_q4_price_indices.csv")
library(readxl); defs    <- read_excel("2022_q4_price_index_data.xlsx",
                                       sheet = "Submarket_Definitions")
* Stata
use price_data.dta, clear
import delimited depaul_q4_price_indices.csv, clear
import excel 2022_q4_price_index_data.xlsx, ///
    sheet("Submarket_Definitions") firstrow clear
```

Reshaping

```
# R
idx_long <- tidyr::pivot_longer(idx, -yearq, names_to = "area", values_to = "index")
* Stata
rename (cook-chicagosouthchicagowestpullman) idx_=
reshape long idx_, i(yearq) j(area) string
```

Descriptives and graphs

```
# R:      mean, sd, quantile, weighted.mean, Hmisc::wtd.quantile
#         ggplot + geom_line / geom_col / geom_density
* Stata: summarize, tabstat [aw=perwt], _pctile
*         graph twoway line, graph bar, kdensity
```

Regression, fixed effects, binary outcomes

```
# R
feols(lnprice ~ sqft2 + numbath | tract + year, data = prices, cluster = ~property_id)
feols(own ~ i(cohort, ref = "boomer") + loginc, data = acs,
      weights = ~perwt, vcov = "hetero")
feglm(own ~ i(cohort, ref = "boomer") + loginc, data = acs,
      weights = ~perwt, family = binomial)
marginaleffects::avg_slopes(model)
predict(model, newdata = profiles, type = "response")
* Stata
reghdfe lnprice sqft2 numbath, absorb(tract year) vce(cluster property_id)
reg own i.cohort loginc [pw=perwt], vce(robust)
logit own i.cohort loginc [pw=perwt]
margins, dydx(*)
```